

WASHINGTON UNIVERSITY IN ST. LOUIS

ADVANCED LINEAR MODELS II

A Hierarchical Occupancy–Abundance–Detection Model for Salamander Counts

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1 Introduction

Modeling species abundance from field data is a central problem in ecological statistics, particularly when observations are collected under imperfect detection. In many ecological surveys, including studies of amphibian populations, the observed data consist of repeated counts across sites and time. These counts often exhibit two prominent features: a high frequency of zeros and substantial variability across locations, arising from both ecological processes (true absence and variation in abundance) and observational processes.

Classical count models, including Poisson regression, are often inadequate due to their restrictive mean–variance relationship. The negative binomial model relaxes this assumption by allowing for overdispersion and is widely used in ecological applications (Hardin & Hilbe, 2007). Zero-inflated models (Lambert, 1992) further accommodate excess zeros through a mixture structure, but they treat zero inflation purely statistically and do not distinguish between ecological absence and nondetection.

Hierarchical models address this limitation by separating the data-generating process into distinct components. In particular, the N-mixture model introduced by Royle (2004) incorporates a latent abundance variable at each site and models observed counts through a detection process conditional on this abundance, enabling inference on both occupancy and true abundance.

In this report, we analyze salamander count data collected across multiple stream sites under varying environmental conditions, including mining activity. The dataset consists of repeated counts of multiple species at each site, making it well suited for an occupancy–abundance–detection framework. We adopt a hierarchical model in which occupancy, latent abundance, and detection are modeled separately, with abundance following a zero-truncated negative binomial distribution to account for overdispersion.

Our primary objective is to quantify the effects of mining and environmental covariates on salamander occupancy and abundance while accounting for imperfect detection. We allow for species-specific effects to capture heterogeneity across species and compare the proposed model with standard zero-inflated negative binomial mixed models fitted using Brooks et al. (2017). By explicitly separating ecological and observational processes, the proposed approach provides a more interpretable framework for understanding species distribution patterns and the impact of environmental disturbances.

2 Data Description

The data analyzed in this study originate from a field investigation of stream salamander populations conducted in eastern Kentucky, USA, as reported in Price, Muncy, Bonner, Drayer, and Barton (2016). The study focuses on understanding how large-scale land-use disturbance, specifically mountaintop removal mining and valley filling (MTR/VF), affects salamander communities in headwater stream ecosystems.

The dataset consists of observations collected from a total of 23 stream sites. Among these, 11 sites are classified as mined streams, located on a reclaimed surface mine where mountaintop removal and valley filling had previously occurred. The remaining 12 sites are reference streams situated in relatively undisturbed forested areas. These two groups of sites were chosen to be comparable in terms of underlying geological and environmental characteristics prior to mining, allowing for meaningful comparison of ecological outcomes.

At each site, salamander populations were surveyed within fixed 10-meter stream reaches. These reaches were selected to ensure suitable habitat conditions, typically including a combination of pools, runs, and riffles. Sampling was conducted repeatedly over time, with each site visited multiple times during the study period. During each visit, researchers performed standardized active searches, including dipnetting within the stream and manual inspection of nearby substrates such as rocks, logs, and leaf litter. All individuals encountered were identified to species, counted, and released.

The dataset includes observations for multiple salamander species, representing both adult and larval life stages. Salamanders are a dominant vertebrate group in these ecosystems and play a key ecological role, making them a useful indicator of environmental health. Their populations are known to be sensitive to habitat

disturbance, water quality, and changes in stream structure.

In addition to salamander counts, each site is characterized by its mining status (mined or non-mined), which serves as the primary explanatory factor in this analysis. Environmental attributes such as habitat structure and water conditions differ substantially between mined and reference sites. For example, mined streams are typically associated with altered vegetation, reduced forest cover, and changes in stream chemistry, all of which can influence salamander populations.

Repeated measurements were collected under varying environmental conditions, including differences in temperature, seasonal timing, and recent weather patterns. These factors influence the detectability of salamanders during surveys and are therefore important for interpreting observed counts.

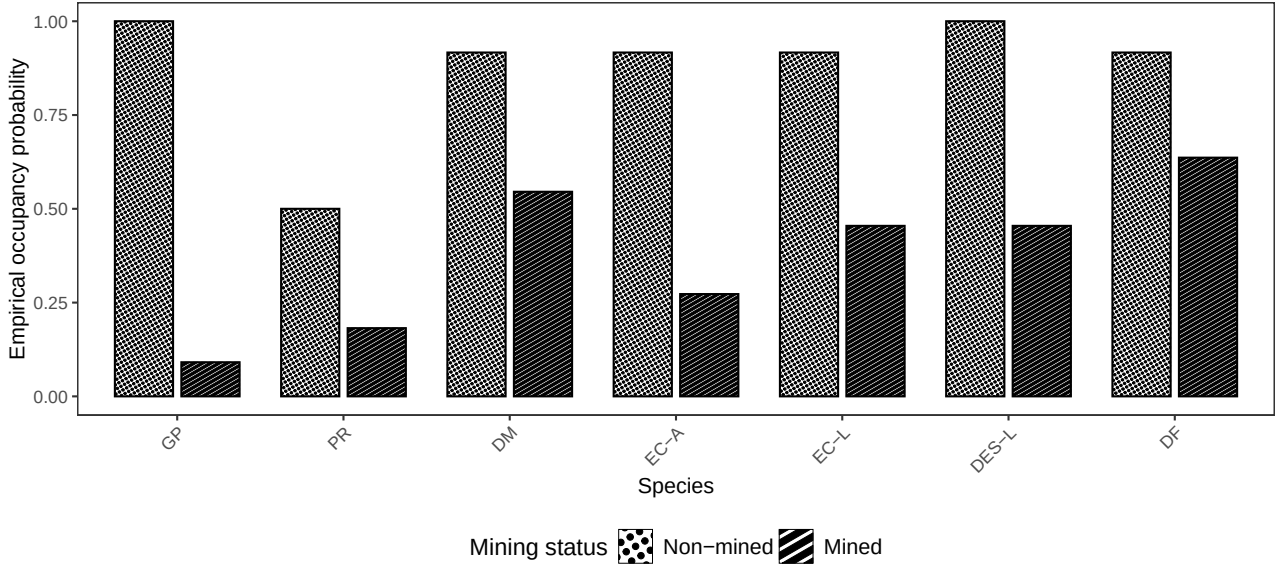


Figure 1: Empirical occupancy probabilities for each species under mined and non-mined conditions.

To provide an initial visual comparison between mined and non-mined sites, we present a summary of empirical occupancy patterns across species in Figure 1. This figure illustrates the proportion of sites at which each species was observed at least once, highlighting systematic differences between disturbed and undisturbed habitats.

Overall, the dataset provides a structured collection of repeated ecological observations across multiple species, sites, and environmental conditions, enabling a detailed investigation of how land-use disturbance influences species presence and abundance in stream ecosystems.

3 Hierarchical Occupancy–Abundance–Capture Model

Since the data contain a large number of zeros—many plausibly arising from environmental disturbance due to mining—it is natural to consider standard zero-inflated count models such as the Poisson or negative binomial. While these models are suitable for count data and computationally convenient, they treat excess zeros purely statistically and do not explicitly reflect the underlying ecological processes.

We instead adopt a hierarchical framework that separates species presence, abundance, and detection, providing a more interpretable representation of the data-generating mechanism.

As shown in Figure 2, salamander counts are generally higher in non-mined sites, with substantial heterogeneity across species. Some species (e.g., DES-L, EC-L, DM, GP) are frequently observed, while others are rarely detected. This raises the question of whether zeros correspond to true absence or imperfect detection, motivating a hierarchical formulation.

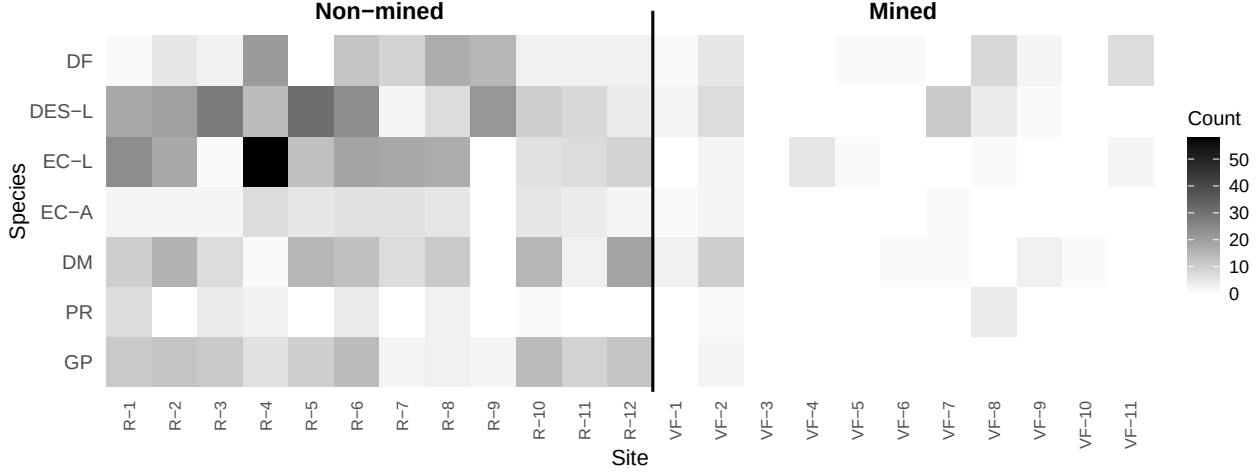


Figure 2: Site-species heatmap of observed salamander counts. Sites are ordered by mining status, with non-mined (R) sites on the left and mined (VF) sites on the right.

Let $i = 1, \dots, I$ index sites, $s = 1, \dots, S$ species, and $j = 1, \dots, J$ sampling occasions. Let Y_{sij} denote observed counts. We introduce a latent indicator

$$Z_{si} \sim \text{Bernoulli}(\psi_{si}), \quad \text{with} \quad \text{logit}(\psi_{si}) = \gamma_{0s} + \gamma_1 \text{mined}_i.$$

This specification is motivated by the empirical summaries in Table 1, which show substantial heterogeneity across species and a consistently negative effect of mining.

Spp	Mined	Occ.		Abund.	
		n_{occ}	$\hat{\psi}$	Total	$\hat{\lambda}$
EC-A	Yes	3	0.2730	4	1.33
	No	11	0.9170	46	4.18
EC-L	Yes	5	0.4550	11	2.20
	No	11	0.9170	190	17.27
DES-L	Yes	5	0.4550	25	5.00
	No	12	1.0000	188	15.67
DF	Yes	7	0.6360	25	3.57
	No	11	0.9170	92	8.36

Spp	Mined	Occ.		Abund.	
		n_{occ}	$\hat{\psi}$	Total	$\hat{\lambda}$
GP	Yes	1	0.0909	2	2.00
	No	12	1.0000	106	8.83
PR	Yes	2	0.1820	5	2.50
	No	6	0.5000	22	3.67
DM	Yes	6	0.5450	19	3.17
	No	11	0.9170	117	10.64

Table 1: Empirical occupancy and abundance summaries by species and mining status.

To examine abundance patterns, we consider observed counts conditional on sites where the species was detected at least once. Define

$$\hat{Z}_{si} = \mathbf{1} \left(\sum_j Y_{sij} > 0 \right),$$

and for each species and mining status,

$$\hat{\lambda}_s^{(m)} = \frac{\sum_{i: \text{mined}_i = m} \sum_j Y_{sij}}{\sum_{i: \text{mined}_i = m} \hat{Z}_{si}}.$$

Table 1 indicates substantial heterogeneity across species even after conditioning on presence, while the

effect of mining remains consistently negative. We model latent abundance as

$$N_{si} \mid Z_{si} = \begin{cases} 0, & Z_{si} = 0, \\ \text{NegBin}^+(\mu_{si}, \phi), & Z_{si} = 1, \end{cases}$$

$$\log(\mu_{si}) = \beta_{0s} + \beta_{1s} \text{mined}_i + \beta_2 \text{cover}_i.$$

The negative binomial distribution accommodates overdispersion commonly observed in ecological count data. Conditional on N_{si} ,

$$Y_{sij} \mid N_{si} \sim \text{Binomial}(N_{si}, p_{sij}),$$

$$\text{logit}(p_{sij}) = \alpha_{0s} + \alpha_{1s} \text{DOP}_{ij} + \alpha_{2s} \text{Wtemp}_{ij} + \alpha_{3s} \text{DOY}_{ij} + \alpha_{4s} \text{Wtemp}_{ij} \text{DOY}_{ij}.$$

Species-specific intercepts α_{0s} allow for differences in baseline detectability. A negative binomial abundance model is a natural extension of the Poisson model here, since ecological abundance data often exhibit overdispersion, with variability exceeding the mean. The additional dispersion parameter ϕ allows the model to capture this extra heterogeneity more flexibly than the Poisson specification. Overall, the model decomposes the data-generating process into occupancy, abundance, and detection components. The full likelihood formulation and its derivation are provided in Appendix A.

The details of parameter estimation are given in Appendix B.

4 Results

Model Comparison

To evaluate the proposed hierarchical occupancy–abundance–detection model, we compare it with a set of standard zero-inflated negative binomial (ZINB) models fitted using the `glmmTMB` framework.

In the first model, salamander counts are modeled as a function of species, mining status, and environmental covariates, with a site-level random intercept to account for dependence among observations from the same site. The zero-inflation component includes both species and mining effects. The second model extends this specification by allowing the effect of mining to vary across species through an interaction between species and mining status. The third model incorporates species-level random effects in both the count and zero-inflation components, while retaining a common mining effect.

These models represent a range of flexible specifications within the ZINB framework. In contrast, the proposed hierarchical model explicitly decomposes the data-generating process into occupancy, abundance, and detection components, thereby providing a structured representation of the underlying ecological mechanisms.

Model	AIC	BIC	logLik
ZINB: spp fixed + mined + covariates + (1 site)	1670.31	1768.60	-813.16
ZINB: spp fixed + spp:mined + covariates + (1 site)	1664.91	1790.00	-804.45
ZINB: spp random + mined + covariates + (1 site)	1688.76	1742.37	-832.38
Proposed hierarchical model (final)	1682.62	1864.43	-782.31

Table 2: Model comparison based on AIC, BIC, and log-likelihood.

The comparison results indicate that the ZINB models achieve strong performance in terms of likelihood-based criteria. In particular, the ZINB model with species-specific mining effects attains the lowest AIC (1664.91).

The proposed hierarchical occupancy–abundance–detection model, after incorporating species-specific detection effects, achieves an AIC of 1682.62 and a log-likelihood of -782.31 . While its AIC is slightly higher than that of the best-performing ZINB model, the hierarchical model attains a substantially higher log-likelihood,

indicating an improved fit to the underlying data-generating process. The higher AIC reflects the penalty for increased model complexity rather than a lack of fit, illustrating the trade-off between parsimony and structural interpretability.

This pattern reflects the fact that ZINB models act as flexible reduced-form representations of the data, directly modeling the observed counts and excess zeros without imposing structural constraints. In contrast, the hierarchical model explicitly separates occupancy, abundance, and detection processes, thereby introducing additional structure.

Importantly, the hierarchical model distinguishes between true absence and imperfect detection, allowing separate inference on species presence, latent abundance, and detection processes. Thus, while the ZINB models achieve slightly better performance under AIC, the proposed hierarchical model offers a more interpretable and ecologically meaningful framework with comparable statistical fit.

Parameter Interpretation

We now interpret the estimated parameters of the hierarchical occupancy–abundance–detection model in relation to the underlying ecological processes.

The species-specific occupancy intercepts γ_{0s} show substantial heterogeneity, indicating differences in baseline presence across species. Species such as DF and DES-L exhibit relatively high baseline occupancy, while species such as PR have comparatively lower occupancy levels. The estimated mining effect $\gamma_1 = -3.106$ is strongly negative, indicating that mining substantially reduces the probability of species presence across all species. On the odds scale, this corresponds to a reduction by a factor of $\exp(-3.106) \approx 0.045$, implying that occupancy odds in mined sites are approximately 4.5% of those in non-mined sites.

The baseline abundance parameters β_{0s} are moderately similar across species, suggesting comparable abundance levels under non-mined conditions. However, the mining effects β_{1s} vary substantially across species and are predominantly negative. For example, GP exhibits a strong decline (-3.03), corresponding to a multiplicative reduction of $\exp(-3.03) \approx 0.048$, while PR shows almost no decline (0.15), indicating minimal sensitivity to mining. This highlights strong species-specific heterogeneity in response to environmental disturbance. The cover effect $\beta_2 = -0.211$ is relatively small, suggesting that vegetation cover plays a limited role compared to mining.

The detection component reveals substantial species-specific variation. All detection intercepts α_{0s} are negative, indicating generally low baseline detectability across species. However, the effects of environmental covariates vary across species. For instance, water temperature has a strong positive effect for EC-A (0.751), while it is negative for EC-L (-0.259), indicating contrasting responses across species. Similarly, seasonal effects (DOY) differ markedly, with EC-A exhibiting a strong negative effect (-2.03) and EC-L a strong positive effect (0.764).

The interaction between water temperature and seasonal timing further refines detection dynamics. For example, EC-A shows a strong positive interaction (0.533), suggesting that detection increases substantially when both temperature and seasonal timing are favorable. In contrast, species such as EC-L and DES-L exhibit negative interaction effects, indicating more complex or opposing seasonal responses.

The estimated dispersion parameter $\log \phi = 0.798$ indicates substantial overdispersion in the count data, justifying the use of a negative binomial specification over a Poisson model.

Overall, the results reveal a strong negative impact of mining on both occupancy and abundance, along with pronounced species-specific variation in detectability and environmental response patterns.

5 Residual Analysis

To assess the adequacy of the fitted hierarchical occupancy–abundance–detection model, we perform simulation-based residual diagnostics using the DHARMA framework.

Occupancy		Abundance		Detection (Intercepts)	
Param	Est	Param	Est	Param	Est
$\gamma_{0,GP}$	2.638	$\beta_{0,GP}$	3.759	$\alpha_{0,GP}$	-2.788
$\gamma_{0,PR}$	1.495	$\beta_{0,PR}$	3.115	$\alpha_{0,PR}$	-3.855
$\gamma_{0,DM}$	3.568	$\beta_{0,DM}$	3.787	$\alpha_{0,DM}$	-2.681
$\gamma_{0,EC-A}$	2.976	$\beta_{0,EC-A}$	2.451	$\alpha_{0,EC-A}$	-3.262
$\gamma_{0,EC-L}$	3.389	$\beta_{0,EC-L}$	3.613	$\alpha_{0,EC-L}$	-2.100
$\gamma_{0,DES-L}$	3.225	$\beta_{0,DES-L}$	3.697	$\alpha_{0,DES-L}$	-1.985
$\gamma_{0,DF}$	4.325	$\beta_{0,DF}$	3.235	$\alpha_{0,DF}$	-2.463
γ_1	-3.106	β_2	-0.211		
Abundance Mining Effects					
$\beta_{1,GP}$	-3.029	$\beta_{1,PR}$	0.152	$\beta_{1,DM}$	-1.493
$\beta_{1,EC-A}$	-2.693	$\beta_{1,EC-L}$	-2.948	$\beta_{1,DES-L}$	-1.518
$\beta_{1,DF}$	-1.139				
Detection Covariates					
$\alpha_{1,DOP,GP}$	-0.008	$\alpha_{2,Wtemp,GP}$	0.236	$\alpha_{3,DOY,GP}$	0.271
$\alpha_{1,DOP,PR}$	0.180	$\alpha_{2,Wtemp,PR}$	0.146	$\alpha_{3,DOY,PR}$	0.258
$\alpha_{1,DOP,DM}$	0.138	$\alpha_{2,Wtemp,DM}$	0.108	$\alpha_{3,DOY,DM}$	0.020
$\alpha_{1,DOP,EC-A}$	-0.204	$\alpha_{2,Wtemp,EC-A}$	0.751	$\alpha_{3,DOY,EC-A}$	-2.031
$\alpha_{1,DOP,EC-L}$	0.043	$\alpha_{2,Wtemp,EC-L}$	-0.259	$\alpha_{3,DOY,EC-L}$	0.764
$\alpha_{1,DOP,DES-L}$	-0.263	$\alpha_{2,Wtemp,DES-L}$	-0.158	$\alpha_{3,DOY,DES-L}$	-0.084
$\alpha_{1,DOP,DF}$	0.154	$\alpha_{2,Wtemp,DF}$	-0.126	$\alpha_{3,DOY,DF}$	0.088
Interaction Effects					
$\alpha_{4,Wtemp:DOY,GP}$	0.054	$\alpha_{4,Wtemp:DOY,PR}$	0.355	$\alpha_{4,Wtemp:DOY,DM}$	0.077
$\alpha_{4,Wtemp:DOY,EC-A}$	0.533	$\alpha_{4,Wtemp:DOY,EC-L}$	-0.194	$\alpha_{4,Wtemp:DOY,DES-L}$	-0.266
$\alpha_{4,Wtemp:DOY,DF}$	0.170				
Dispersion					
$\log \phi$	0.798				

Table 3: Estimated parameters of the hierarchical occupancy–abundance–detection model.

Figure 3 summarizes the diagnostic results. The QQ plot of residuals indicates no significant deviation from the expected uniform distribution (Kolmogorov–Smirnov test, $p = 0.239$), suggesting that the overall distributional assumptions of the model are appropriate. The dispersion test does not detect evidence of overdispersion or underdispersion ($p = 0.436$), indicating that the negative binomial specification adequately captures variability in the count data. Similarly, the zero-inflation test yields a non-significant result ($p = 0.08$), suggesting that the model sufficiently accounts for the observed frequency of zeros without requiring additional zero-inflation structure.

The outlier test is also non-significant, indicating no extreme observations that are poorly explained by the model. Furthermore, the residuals plotted against predicted values do not exhibit systematic patterns, supporting the absence of model misspecification in the mean structure.

Overall, these diagnostics provide no evidence against the fitted model, indicating that it offers an adequate representation of the underlying data-generating process.

6 Discussion

The results of the hierarchical occupancy–abundance–detection model underscore the importance of separating ecological and observational processes when analyzing count data under imperfect detection. Unlike standard zero-inflated models, the proposed framework distinguishes between true absence and nondetection, enabling direct inference on species occupancy and latent abundance.

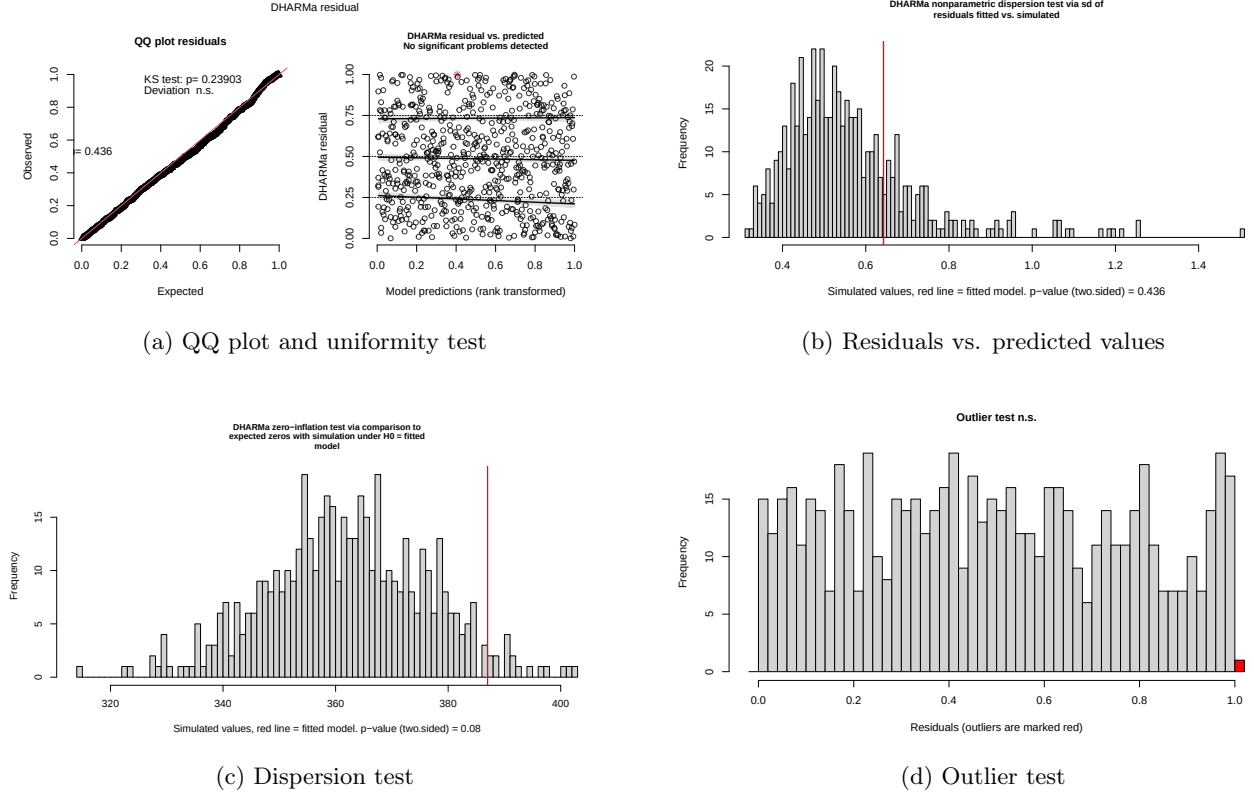


Figure 3: Simulation-based residual diagnostics for the final hierarchical occupancy–abundance–detection model.

The estimated parameters indicate a strong negative effect of mining on both occupancy and abundance. The occupancy effect implies a substantial reduction in the probability of species presence at mined sites, while species-specific abundance effects reveal heterogeneous responses, with some species exhibiting pronounced declines and others showing weaker sensitivity.

The detection component shows considerable heterogeneity across species and environmental conditions. Baseline detection probabilities are generally low, and environmental covariates such as water temperature and seasonal timing have species-specific effects, including notable interactions. These results highlight the importance of accounting for detection when interpreting observed counts.

Model comparison indicates that zero-inflated negative binomial models achieve slightly better AIC, reflecting their flexibility as reduced-form models. However, the hierarchical model attains a higher log-likelihood, suggesting a better fit to the underlying data-generating process. The higher AIC reflects the additional parameters required to model occupancy, abundance, and detection separately, illustrating the trade-off between parsimony and interpretability.

Residual diagnostics do not reveal major model misspecification. The absence of significant deviations in uniformity, dispersion, and zero-inflation tests suggests that the hierarchical structure and negative binomial assumption adequately capture key features of the data.

Despite these strengths, the model assumes independence across species and sites and relies on species-specific parameters, which may lead to overparameterization in limited samples. Future work could incorporate random effects or hierarchical shrinkage to improve stability.

Overall, the proposed framework provides a principled and interpretable approach to modeling ecological count data under imperfect detection, offering insights into occupancy, abundance, and detectability beyond conventional models.

References

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A Likelihood Formulation

For a fixed species–site pair (s, i) , let

$$\mathbf{Y}_{si} = (Y_{si1}, \dots, Y_{siJ_i})$$

denote the vector of observed counts across the J_i sampling occasions.

Under the proposed hierarchical model, the occupancy indicator follows

$$Z_{si} \sim \text{Bernoulli}(\psi_{si}),$$

and conditional on occupancy, the latent abundance is modeled as

$$N_{si} \mid Z_{si} = \begin{cases} 0, & Z_{si} = 0, \\ \text{NegBin}^+(\mu_{si}, \phi), & Z_{si} = 1, \end{cases}$$

where $\text{NegBin}^+(\mu_{si}, \phi)$ denotes the zero-truncated negative binomial distribution.

Given the latent abundance $N_{si} = n$, the observed counts across sampling occasions are assumed conditionally independent, with

$$Y_{sij} \mid N_{si} = n \sim \text{Binomial}(n, p_{sij}), \quad j = 1, \dots, J_i.$$

The likelihood contribution for the (s, i) th species–site pair is obtained by marginalizing over the unobserved abundance N_{si} .

Case 1: No detections. If no individuals are observed across all sampling occasions, that is, if $y_{si1} = \dots = y_{siJ_i} = 0$, then two scenarios are possible: either the species is truly absent, or it is present but all individuals go undetected. Accordingly, the likelihood contribution is given by

$$L_{si} = (1 - \psi_{si}) + \psi_{si} \sum_{n=1}^{\infty} \frac{P_{\text{NB}}(n; \mu_{si}, \phi)}{1 - P_{\text{NB}}(0; \mu_{si}, \phi)} \prod_{j=1}^{J_i} (1 - p_{sij})^n.$$

Case 2: At least one detection. If at least one positive count is observed, that is, if $\max_j y_{sij} > 0$, then the species must be present at the site. In this case, the likelihood is obtained by summing over all feasible values of the latent abundance that are consistent with the observed data. Specifically,

$$L_{si} = \psi_{si} \sum_{n=\max_j y_{sij}}^{\infty} \frac{P_{\text{NB}}(n; \mu_{si}, \phi)}{1 - P_{\text{NB}}(0; \mu_{si}, \phi)} \prod_{j=1}^{J_i} \binom{n}{y_{sij}} p_{sij}^{y_{sij}} (1 - p_{sij})^{n-y_{sij}}.$$

Assuming independence across species–site pairs, the full likelihood is

$$L(\theta) = \prod_{s=1}^S \prod_{i=1}^I L_{si},$$

and the corresponding log-likelihood is

$$\ell(\theta) = \sum_{s=1}^S \sum_{i=1}^I \log L_{si}.$$

This likelihood forms the basis for parameter estimation.

B Parameter Estimation

The model parameters are estimated by maximizing the log-likelihood described in Appendix A. Because the abundance variable N_{si} is latent, the likelihood contribution for each species–site pair involves summation over all feasible values of N_{si} . This infinite sum does not admit a closed-form expression and is therefore evaluated numerically.

For each species–site pair, the summation is truncated at a finite upper bound. Specifically, we sum over $n = \max_j y_{sij}, \dots, K$, where $K = \max_j y_{sij} + K_{\text{extra}}$. In the implementation, we set $K_{\text{extra}} = 150$, which provides a sufficiently accurate approximation to the infinite sum in all cases considered.

All likelihood calculations are carried out on the log scale, and the summation over latent abundance is evaluated using a log-sum-exp stabilization to ensure numerical stability. The dispersion parameter of the negative binomial distribution is estimated on the log scale by optimizing over $\log \phi$ and setting $\phi = \exp(\log \phi)$, thereby enforcing the positivity constraint.

Initial values are constructed using empirical summaries. Species-specific occupancy intercepts are initialized from observed occupancy probabilities in non-mined sites. Species-specific abundance intercepts and mining effects are initialized using empirical abundance summaries across mined and non-mined sites. Detection parameters, including the species-specific regression coefficients and interaction terms, are initialized at neutral values.

The resulting high-dimensional optimization problem is solved using the `nlmminb` routine in **R**. Convergence is assessed based on the optimizer’s convergence code, stability of the objective function, and the absence of extreme or implausible parameter estimates.